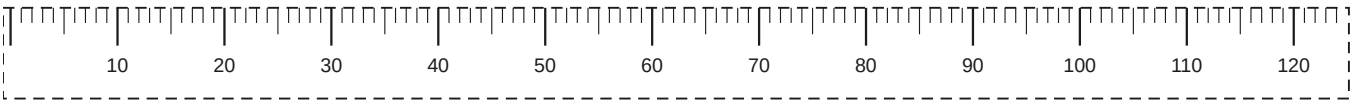
# Bench exercise: make a back scale and read a log

Japanese Joinery by Hand · Episode 2 · Part 1. You don't need a sashigane: you'll make a paper *ura-me* 裏目 (√2 scale) and check, with your own hands, that it reads the largest square in a circle. About 30 minutes.

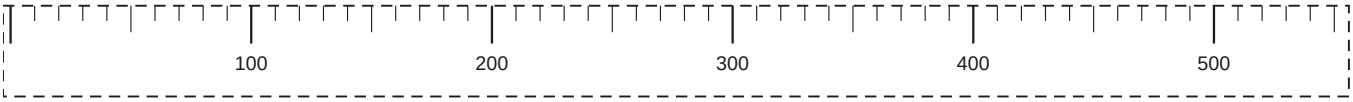
> **Print at 100% / "actual size"** (not "fit to page"). Then check the 100 mm calibration bar with a ruler. If it isn't 100 mm, the scales below are wrong; reprint before using them.

**calibration: 100 mm**

**back scale (ura-me): 1 unit = 1.414 mm. Cut out along the border.**




**round scale (marume): 1 unit = 1/π mm = 0.318 mm (ticks every 10 units)**




## 1 · Prove the √2 on paper

1. Draw a square 100 mm on a side, with a good square and a sharp pencil.
2. Measure its diagonal with an ordinary ruler. You should get about 141 mm (100 × 1.414).
3. Now lay the **back scale** strip along the same diagonal. It reads **100**: the side.

## 2 · Read a "log"

1. Find something round with a flat end: a log or branch end, a dowel, a jar lid, a pipe.
2. Lay the back scale across the end, **straight through the center**, zero on one edge. Read it at the far edge. That number is the side of the largest square you could cut from it.
3. For a real log, measure at the **small end**, since a log tapers.

## 3 · Check it

1. Trace the round end on paper. Draw two diameters at right angles through the center (fold the paper, or use a square).
2. Join the four points where they meet the circle. That's the largest inscribed square. Measure its side with an ordinary ruler: it should match your back-scale reading, within about 1 mm.
3. Try a square 1 mm bigger: its corners poke outside the circle.

## 4 · Optional: the round scale

- Lay the **marume** strip across the same diameter and read it. Then wrap a string around the object and measure the string with an ordinary ruler. The two numbers should agree: the round scale reads the circumference.

## What you just did

A square's diagonal is its side × √2. A scale stretched by √2 turns any diagonal straight into the side, with no arithmetic. Across a circle, the diameter *is* the inscribed square's diagonal. That's all the sashigane's back scale is doing.

The strips are for learning, not precision work: paper stretches and printers vary. On a real sashigane, scale placement varies by model; one common layout has the ura-me on the back of the long arm's outer edge and the marume on the back of the short arm.